

facility, ensuring high-quality control and the ability to customise quickly. This strategy also protects their intellectual property. Ajith emphasises that “the machine, which offers a resolution of $\pm 0.01\text{mm}$ and placement accuracy of $\pm 0.05\text{mm}$, is ideal for small-scale and batch production.” The company has expanded its product line to include reflow ovens and solder paste dispensers, catering to the diverse needs of modern electronics manufacturers.

The need for a prototyping solution emerged in 2017 due to inefficiencies and high costs associated with manual handling and reliance on external assembly houses, which were unsuitable for experimentation. This led to the decision to build a pick-and-place machine, initially underestimated in complexity but later revealed to demand significant precision-related problem-solving. By 2019-2020, after overcoming numerous technical challenges, the first machine version was developed. However, the Covid-19 pandemic introduced new obstacles, including a critical chip shortage that forced a re-evaluation of design and

components. These challenges, along with internal strategic shifts, led to a regrouping under the ‘Xaltan Systems’ banner and a transition to an open source software base, ultimately resolving many of the difficulties encountered during development.

The machines deliver performance with good resolution, placement accuracy, and speeds up to 3000 CPH, making them ideal for small-scale and batch production. Integrating advanced hardware and software, they feature a custom vision system and flexible feeders to manage various component sizes and orientations efficiently. In 2020, the company expanded its product line to include reflow ovens and solder paste dispensers, further enhancing its reputation. These machines are now recognised for their precision, versatility, and quick changeover capabilities, redefining desktop PCB assembly.

Committed to sustainability, they adhere to the restriction of hazardous substances (RoHS) directive, ensuring environmentally friendly manufacturing processes. Their energy-efficient

machines consume minimal power, supporting sustainable practices. The company has received substantial support from the Ministry of Electronics and Information Technology (MeitY), which has been instrumental in their product development and market expansion efforts. The Technology Development Board (TDB) had also recognised this innovation with an award in 2022. Maker Village, Kochi, and the Kerala Startup Mission (KSUM) have also supported and encouraged the effort all through.

The startup aims to expand its product line to handle more complex production requirements and higher volumes. It is exploring international markets and seeking partnerships with distributors, component suppliers, and academic institutions. The company is particularly interested in channel partners with strong technical expertise in electronics manufacturing for global distribution. While currently not engaged in academic collaborations, it remains open to such partnerships to bolster its R&D efforts and continue driving innovation in its field.

—Akanksha Sondhi Gaur

3 InsideFPV DRONE TECHNOLOGY FOR AGRICULTURE, DEFENCE, AND ACCESSIBLE CONSUMER SOLUTIONS

InsideFPV Private Limited, a deep-tech drone company, was founded in July 2020 amid the Galwan Valley conflict between India and China.

Founded by Arth Chowdhary, Deyvant Bhardwaj, and Oshi Kumari, the company was born from India’s technological challenges at the time of Galwan Valley conflict in 2020. Its mission is to strengthen the nation’s defence and security through drone



technology. With backgrounds in UAVs, aircraft, and drone systems, the founders had previously worked with key government organisations on various projects. Their combined expertise formed the bedrock of InsideFPV.

Explaining the name InsideFPV, Arth Chowdhary explains that FPV stands for ‘first-person view,’ embodying the company’s mission to create technology that enables the remote control of